

IC 222P: Physics Practicum/Practicals

Course Code: IC 222P

Physics Practicum/Practicals

L-T-P-C: 0-0-3-2

Prerequisites: Consent of the faculty member

Students intended for: B.Tech

Elective or Core: Core

Approval: 2nd Senate

Practicum kind of experiments

1. Fourier series: Observing Fourier series in real life (simple electronic circuits) by making suitable arrangements.
2. Four Probe method: Temperature dependent resistivity of a semiconductor, finding the Band gap.
3. Newtons Ring: Division of amplitude, Interference, wavelength of source
4. Fresnel biprism: Division of wavefront, Interference, wavelength of source
5. Fraunhofer Diffraction: Study the diffraction effects by designing suitable slits (single/double)
6. Dielectric properties of material: Determination of dielectric constant of glass, wood
7. Make capacitor and measure the charging and discharging of the capacitor using different dielectric materials, various thickness. To generate potentials of different shapes and study the motion of the body in or through them

Standard experiments

1. Mechanical Hysteresis: Relationship between torque and rotation of a metal bar (steel, aluminium, brass, Copper), Observation of memory effect (elasticity, plasticity, relaxation).
2. Frank Hertz experiment: To study the excitation potential of a gas molecule.
3. Magnetron method: Charge to mass ratio of an electron
4. Magnetic field due to a single coil : Magnetic field along the axis of the coil at different positions, Effect of different coil radius
5. Hemholtz coil: Magnetic field for different separation of the coils, Superposition of field.
6. Magnetic Induction: Measure induced emf as a function of rate of change of flux.
7. Millikan's oil drop experiment: Determine the elementary charge
8. Coupled Oscillator: Coupled vibration, Beats, Coupling of energy between two harmonic oscillators coupled to each other.